

ID No. _____

Birla Vishvakarma Mahavidyalaya (Engineering College)
(An Autonomous Institution)
Third Year, B.Tech.(Computer Engineering)
End Semester Examination, Summer - 2023 (AY 2022-23)

Course Code: 3CP07 Course Title: Software Engineering

Date: 26.04.2023

Time: 11:00 am to 01:30 pm

Maximum Marks: 60

Instructions:

- Each section must be written in a separate answer book.
- Numbers written in the square brackets to the right indicate maximum marks.
- The text just below marks indicates the Course Outcome Nos. (CO) followed by the Bloom's taxonomy level of the question, i.e., R: Remember, U: Understand, A: Apply, N: Analyze, E: Evaluate, C: Create

Section – I

- Q. 1 (a) Discuss why software doesn't wear out in comparison to hardware with a suitable diagram. [04]
1U
- (b) Software engineering is a layered technology. Explain in details. [04]
1U

OR

- (b) Draw a diagram of a generic process framework. Explain the framework activities carried out during the software development process. [04]
1,5 U
- (c) Explain the workings of the incremental model and list its advantages. [03]
1U
- Q. 2 (a) List out project factors that should be considered when planning the structure of software teams. Explain why software teams suffer from team toxicity. [05]
5R
- (b) Define DRE. Explain size-oriented metrics for various ongoing projects in organization. [04]
5R
- Q. 3 (a) Answer the following questions (**Any Two**) [06]
1,5U
1. Explain organizational paradigm for software engineering team.
 2. Discuss process metrics guidelines to measure software process.
 3. What is the importance of software estimation? List out different factor on which estimation accuracy depends.
- (b) What is SRS? Explain characteristics of good SRS. Why SRS is known as black-box specification of the system? [04]
2R

Section – II

- Q. 4 (a) List different resource required to develop software projects. Explain software resource categories that should be considered as planning proceeds. [05]
5R
- (b) What is software project scheduling? Explain the relationship between effort and delivery time using PNR curve. [04]
5U

- Q. 5 (a) List out fundamental concepts of software design. Explain abstraction in details. [04]
5R
- (b) Explain user interface design process steps by using a suitable diagram. [04]
5U

OR

- (b) Explain Earned Value Analysis (EVA) in details. [04]
5U
- Q. 6 (a) Explain software testing strategies with suitable diagram. [05]
4U
- (b) What is the purpose of software testing? Differentiate white-box testing and black-box testing. [03]
4N

OR

- (b) Define risk projection? List out risk projection steps. [03]
6R
- (c) Consider an eLearning platform where the student or learner can register and browse different courses and join same. They can buy membership of platforms or enroll course by paying fees using various payment methods for selected courses. Talk or chat with an expert related to any query. They can refer online resources like books, journals, magazines, newspapers etc. and can purchase a hard copy of same. They can discuss on specific topics in the discussion forum between joined learners. After successfully completing the course and submitting quiz and assignments, they will get a certificate of completion according to policy and terms. Depict scenario-based modeling using a use case diagram. [05]
3C